

IIT Roorkee
Technical Communication Lab
Question 1 (10 Marks)

Algorithm Focus: Dijkstra's Shortest Path Algorithm

Important Instructions:

- You must write your answer in $\text{\LaTeX}$.
- For the algorithm, use the packages:

`algorithm` and `algpseudocode`

- For the table, use a proper $\text{\LaTeX}$ table environment. You may use `tabular`; use of `booktabs` is recommended.
- The graph is **undirected** and all edge weights are non-negative.
- If multiple nodes have the same tentative distance, choose the **lexicographically smaller** node.
- Process neighbors in **alphabetical order**.
- You must reproduce the complete algorithm and table in $\text{\LaTeX}$ with all blanks filled correctly.

Question:

Write the following in $\text{\LaTeX}$:

a) **Algorithm in $\text{\LaTeX}$ with Filled Blanks (3 Marks)**

Using the `algorithm` and `algpseudocode` packages, write the complete pseudocode of Dijkstra's Algorithm by reproducing the following structure and filling in all blanks correctly:

For each vertex $v \in V$:

$\text{dist}[v] \leftarrow \underline{\hspace{2cm}}, \quad \text{visited}[v] \leftarrow \underline{\hspace{2cm}}$

Then set:

$\text{dist}[s] \leftarrow \underline{\hspace{2cm}}$

Repeat for $|V|$ iterations:

- choose the unvisited vertex u with minimum $\underline{\hspace{2cm}}$

- if $\text{dist}[u] = \underline{\hspace{2cm}}$, stop
- set $\text{visited}[u] \leftarrow \underline{\hspace{2cm}}$
- for each neighbor v of u in alphabetical order:
 - if $\text{visited}[v] = \underline{\hspace{2cm}}$ and $\text{dist}[u] + w(u, v) < \underline{\hspace{2cm}}$
 - then set $\text{dist}[v] \leftarrow \underline{\hspace{2cm}}$

Return $\underline{\hspace{2cm}}$.

Your answer must be written as a properly formatted $\text{\LaTeX}$ algorithm, not as plain text.

b) **Execution Table in $\text{\LaTeX}$ with Filled Entries (5 Marks)**

Given the graph:

Edge	Weight
A–B	4
A–C	2
B–C	1
B–D	5
C–D	8
C–E	10
D–E	2
D–F	6
E–F	3

Source node: **A**

Using the same tie-breaking and neighbor-order rules, create the following table in $\text{\LaTeX}$ and fill all rows and columns correctly:

Step	Finalized Node	Distance Vector (A, B, C, D, E, F)
0	–	$(0, \infty, \infty, \infty, \infty, \infty)$
1	$\underline{\hspace{2cm}}$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
2	$\underline{\hspace{2cm}}$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
3	$\underline{\hspace{2cm}}$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
4	$\underline{\hspace{2cm}}$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
5	$\underline{\hspace{2cm}}$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}})$
6	$\underline{\hspace{2cm}}$	$(\underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}}, \underline{\hspace{1cm}})$

Your answer must contain the full $\text{\LaTeX}$ code for the table.

c) **Final Result in L^AT_EX (1 Mark)**

Write the following two lines in L^AT_EX and fill in the correct values:

Shortest distance from A to F = _____

Shortest path from A to F = _____

d) **Precision Statement in L^AT_EX (1 Mark)**

Write the following sentence in L^AT_EX and complete it correctly in **40–50 words**:

In Dijkstra's Algorithm, the relaxation step checks whether _____.

If the condition _____ holds, the algorithm updates _____.

Marking Scheme Summary:

- Part (a): 3 marks
- Part (b): 5 marks
- Part (c): 1 mark
- Part (d): 1 mark

Required L^AT_EX packages for the answer:

```
\usepackage{algorithm}
\usepackage{algpseudocode}
\usepackage{booktabs}
```

2. [10 marks] Write the $\text{\LaTeX}$ code for the following.

In the following, we show how to use the `minipage` environment to combine text and logic circuit diagrams as shown in fig.1.

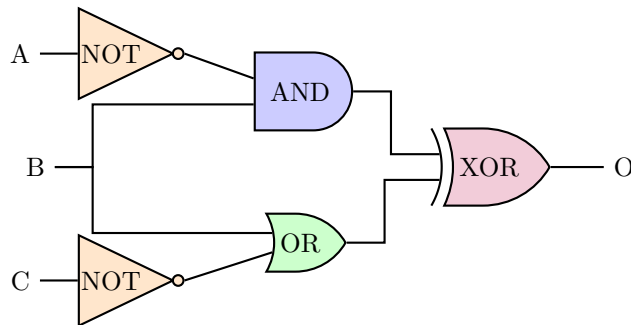


Figure 1: Combinational Logic Circuit

Digital Logic Design

The logic circuit shown in the image is a combinational system that processes three binary inputs (A, B, and C) through three stages of logic to determine the final output O. In the first stage, inputs A and C are passed through NOT gates to create their inverted complements, $\neg A$ and $\neg C$. Next, an AND gate combines $\neg A$ with input B, while an OR gate combines input B with $\neg C$. Finally, these two intermediate results are fed into an XOR (Exclusive OR) gate, which produces a "True" output only if the signals from the AND and OR stages are different. The overall behavior can be summarized by the Boolean expression $O = (\neg A \wedge B) \oplus (B \vee \neg C)$.